

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103
Summer Semester Examination –2023**

Branch: Electronics and Telecommunication Engineering

Sem.:- VI

Subject with Subject Code: - Digital Communication BTETC602

Marks: 60

Date:-14/07/2023

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. **ALL** questions are compulsory.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

	(Level)	(Marks)
Q.1 Solve any two questions from the followings:		
A) Explain Adaptive Delta Modulation with neat diagram. Differentiate between Delta Modulation and Adaptive Delta Modulation.	L1	(06)
B) A channel with bit rate $R_b = 36$ kbps is available for PCM voice transmission. Find appropriate values of binary digits N , the number of quantization levels M and the sampling rate f_s , assuming $f_m = 3.2$ kHz.	L2	(06)
C) Differentiate between Analog and Digital Communication.	L1	(06)
Q.2 Solve any two questions from the followings:		
A) Explain Narrow band noise. Explain its Representation in terms of in phase & quadrature components.	L2	(06)
B) A Television signal with a bandwidth of 4.2 MHz is transmitted using binary PCM. The number of quantization levels is 512. Calculate: 1. Code word length 2. Transmission bandwidth 3. Final bit rate 4. Output signal to quantization noise ratio	L3	(06)
C) Explain the working of Additive Scramblers.	L1	(06)

Q.3 Solve any two questions from the followings:

- A) Write a note on Ergodic Processes. L3 (06)
- B) With neat waveforms explain the concept of ISI digital communication? L2 (06)
- C) Explain various Data encoding or Line coding techniques. L1 (06)

Q.4 Solve any two questions from the followings:

- A) Explain Binary Phase Shift Keying technique with neat waveforms and block diagram. What is the probability of error in BPSK. L2 (06)
- B) Consider an LTI system with impulse response $h(t)$. Let $X(t)$ be a WSS random process. If $X(t)$ is the input of the system, then the output, $Y(t)$, is also a random process. If $Y(t) = \int_{-\infty}^{\infty} h(u)X(t-u)du$, then Prove that : L3 (06)
 - 1. $R_{xy}(\tau) = R_{xx}(\tau) * h(-\tau)$ and
 - 2. $R_{yy}(\tau) = R_{xy}(\tau) * h(\tau)$ where $*$ denotes convolution
 - 3. $S_{xy}(w) = S_{xx}(w)H^*(w)$ and
 - 4. $S_{yy}(w) = S_{xx}(w)|H(w)|^2$
- C) Explain the working of OFDM technique. What are its advantages? L1 (06)

Q.5 Solve any two questions from the followings:

- A) What is matched filter? Explain its working. L2 (06)
- B) Explain Jamming. What are the various types of Jammers? L2 (06)
- C) Explain the working of DS-SS Scheme with neat block diagram. L2 (06)